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Changing the criteria applied to acceleration and deceleration efforts changes the types of player actions detected

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ABSTRACT

This study identified and profiled the types of player actions of Australian football players executed during high acceleration and/or deceleration efforts and identified how the duration exceeding the acceleration or deceleration threshold influenced the identified executed player actions. The commencing speed of the acceleration and deceleration efforts were also assessed. Acceleration and deceleration efforts of 22 Australian football players monitored with a 10-Hz GNSS device across two matches were analysed. Synchronised video footage of the matches was used to identify the player actions executed during acceleration and deceleration efforts. Linear mixed models were conducted to examine the relationship between commencing speed, player action, and duration exceeding the effort threshold. 'Linear movement' and 'jump' were the most frequently executed player actions for high accelerations and 'stop' and 'being tackled' for high decelerations. Significant differences were observed in commencing speed between player actions for the high deceleration efforts. Furthermore, it was found that a higher commencing speed was associated with an increased duration exceeding the deceleration threshold. These findings will help practitioners and researchers with selecting minimum-duration criteria for acceleration and deceleration efforts and provide an understanding of the types of player actions executed during these efforts.

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KEYWORDS
GPS; athlete tracking system; football; effort criteria

Introduction

Athlete tracking technologies enable the quantification of a player's physical movement during training and competition by measuring their distance travelled, speed, and acceleration (Torres-Ronda et al. 2022). It is common in professional Australian Football (AF) to use Global Positioning System (GPS) derived acceleration data to measure physical movement (Johnston et al. 2018). In current practice, one variable used to quantify physical movement is the number of discrete acceleration and deceleration (negative acceleration values) efforts undertaken (Ellens et al. 2022), which are quantified by categorising instantaneous acceleration data. Researchers and practitioners are interested in quantifying high acceleration and deceleration efforts (e.g., $>\pm 2\ m \cdot s^{-2}$, $>\pm 3\ m \cdot s^{-2}$) because high efforts are important to achieve key match events, such as outrunning an opponent to find space or chasing an opponent (Gray and Jenkins 2010). High acceleration efforts are also considered more energetically demanding than low efforts (Osgnach et al. 2009). The number of acceleration and deceleration efforts performed is used to monitor training load, profile physical movement, and link these measures to injury (Akenhead and Nassis 2016; Harper et al. 2019). Efforts are defined using set criteria such as the acceleration threshold and minimum duration above the threshold (also known as dwell time), which may vary between users. The threshold of an effort defines an acceleration or deceleration category (e.g., $2–3\ m \cdot s^{-2}$, $>\pm 3\ m \cdot s^{-2}$), which are often associated with qualitative descriptors referring to the magnitude of the acceleration/deceleration, e.g., low, medium, and high (Delaney et al. 2018; Oliva-Lozano et al. 2020). The minimum duration is the length of time a player must exceed a set threshold for an effort to be counted (Varley et al. 2017). These criteria ensure spikes in the data due to device measurement error are not detected as efforts, and avoid the detection of multiple efforts where only one occurred, as acceleration data could oscillate around a threshold. The criteria applied to the acceleration/deceleration data can be selected according to the manufacturer's software or custom analysis by researchers (Varley et al. 2014; Akenhead et al. 2016; Abbott et al. 2018). There is currently a lack of consensus in selecting these criteria, resulting in variable and inconsistent criteria within studies of the same sport (Sweeting et al. 2017).

High acceleration and deceleration efforts have also been quantified based on the commencing speed of the effort (Varley and Aughey 2013; Mara et al. 2017). It has been observed that the majority of high acceleration and deceleration efforts occurred from a low commencing speed $<3\ m \cdot s^{-1}$ (Varley and Aughey 2013; Mara et al. 2017), which could be explained by the force–velocity relationship. At lower speeds, players can apply more force and accelerate, whereas at higher speeds they can apply less force and it becomes harder to accelerate (Morin et al. 2011). Furthermore, different energy costs and metabolic power output have been associated with different speeds of the acceleration and deceleration efforts (Osgnach et al. 2009). A speed of $2.5\ m \cdot s^{-1}$ with an acceleration of $3.5\ m \cdot s^{-2}$ has been associated with a metabolic power of

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approximately 51 W · kg^{-1}, whereas a deceleration of −3.5 m · s^{-2} at the same speed has been associated with a metabolic power of approximately 11 W · kg^{-1} (Osgnach et al. 2009). The investigation of acceleration and deceleration efforts in conjunction with their commencing speed yields a better understanding of the effort variable and provides more context.

High acceleration and deceleration efforts in AF are typically presented in research as the number of efforts performed (Harper et al. 2019; Kelly et al. 2019; Rennie et al. 2020). Significant differences exist in the reported number of efforts between positional groups (Johnston et al. 2015), types of competition (Kelly et al. 2019) and phases of play (Rennie et al. 2020). This information is important for practitioners to prescribe training that replicates the overall characteristics required during match play. However, these findings provide little context for how a player achieved a given high acceleration or deceleration effort. A player can theoretically perform the same amount of high acceleration and deceleration efforts but achieve these in different ways. For example, a high acceleration effort could be considered to occur incidentally (e.g., when a player falls), or purposefully (e.g., when a player accelerates in a linear fashion to chase an opponent). Deceleration efforts could occur incidentally, e.g., when a player is tackled, and purposefully, e.g., when a player voluntarily slows down. Furthermore, an acceleration or deceleration effort is often perceived and trained as a running-related player action but may occur during non-running-related player actions such as performing a jump or getting tackled. Currently, there is no delineation between these types of acceleration and deceleration efforts, making it unclear what player actions correspond to the efforts. Therefore, identifying the specific player actions that correspond to high acceleration and deceleration efforts and the frequency that they occur would provide a more thorough evaluation of the acceleration and deceleration effort variable in sport and could increase the specificity of training practices.

The selected minimum duration criteria have been shown to influence the number of detected acceleration efforts (Varley et al. 2017), where an increase in minimum duration was linked to an exponential decline in the number of detected acceleration efforts. As the number of detected efforts differs based on the selected minimum duration, the specific player actions corresponding to these efforts might vary. Previous research has shown that player actions such as 'cutting from a split step' were associated with lower acceleration values than player actions such as tackles (MacLeod et al. 2018; Nagano et al. 2018). Identifying what duration exceeding the high threshold corresponds to which player action will help practitioners and researchers with selecting the minimum duration criteria to collect data (player actions) they are most interested in. Furthermore, it will allow practitioners and researchers to understand how players accumulate high acceleration and deceleration efforts with different minimum durations. Therefore, the primary objective of this study was to identify and profile the types of player actions of Australian Football players executed during high acceleration and/or deceleration efforts and identify what duration exceeding the high threshold corresponds to these player actions. A secondary aim is to investigate the commencing speed profile of the acceleration and deceleration efforts.

Methods

The study sample consisted of 22 male elite AF players from one Australian Football League team. Player movement data collected with a 10-Hz GNSS tracking technology device from two matches during the 2019 competitive season were analysed. Synchronised video footage of the matches was used to analyse the executed player actions during acceleration and deceleration efforts. The procedures used in this study were conducted with approval from the Human Research Ethics Committee of La Trobe University (reference number: HEC21282).

An athlete tracking technology device, consisting of 10-Hz GNSS (Vector S7, Catapult Innovations, Melbourne, Australia), was used to measure player movement, including measures of acceleration, speed, and distance. The device was placed between the athlete's scapulae using the manufacturer's garment with a tight fit, to avoid unnecessary device movement. Data collection procedures were in accordance with the guidelines of Malone et al. (2017), with each athlete having their own specific device. The study sample had an average (± SD) horizontal dilution of precision (HDOP) of 0.60 ± 0.06 and 13 ± 1 number of satellites.

The Vector S7 has reported acceptable levels of validity and reliability for distance (mean bias ≤1%, CV ≤ 1.3%) and validity for speed (mean bias −0.5%) by Catapult's vector data integrity testing (Catapult 2020). To the authors' knowledge, there are no peer-reviewed published data on the validity and reliability of the Catapult Vector system. Data were downloaded using the manufacturer's software (Openfield, version 2.7.1, Catapult Sports, Melbourne, Australia). The raw GNSS speed data was calculated using the Doppler-shift method and was exported for custom analysis. The speed measure was filtered with a 4th order, low pass Butterworth filter with a 1.5 Hz cut-off frequency. Residual analysis was performed, as previously described (Winter 2009), to examine the differences between raw and filtered data over a range of cut-off frequencies (0.1 to 10 Hz in steps of 0.1 Hz) which resulted in an optimal cut-off frequency of 1.5 Hz. Acceleration data were derived from the filtered speed using the forward difference method. Filtering the speed data before deriving the acceleration data will ensure that any noise present in the speed data will not be increased due to deriving. It was chosen to derive the acceleration data rather than using the acceleration data of the manufacturer's software, to create a generic study anyone can replicate (not solely based on the used tracking system). All data were analysed across the full match duration, including any stoppage time.

Timestamps of the start and end of the high acceleration and deceleration efforts were extracted from the acceleration data. The start of a high effort was defined by a threshold of ±3 m · s^{-2} (positive threshold defines an acceleration effort and a negative threshold a deceleration effort). This threshold was selected as it is commonly used for high acceleration and deceleration efforts in the research literature (Delaney et al. 2018; Oliva-Lozano et al. 2020). An effort was counted when the acceleration data reached the set threshold and ended with the acceleration data reaching 0 m · s^{-2} (Varley et al. 2017). The duration exceeding the

high threshold was extracted for each detected effort. The durations were multiples of 0.1 seconds, as this was the sample rate of the collected data. Six other different physical movement measures were calculated for each effort: two acceleration measures (average and max), three speed measures (average, max, commencing speed), and total duration. There were a total of 1178 detected acceleration and deceleration efforts, with a mean ± SD of 46 ± 8 efforts per player per match. All data were processed in MATLAB software unless stated otherwise.

Four time-synchronised video files for each match (side on and behind the goal perspective) were cross-referenced with tracking data and used to determine the executed player action at the extracted acceleration and deceleration effort timestamp. A review of the literature (Dawson et al. 2004; Robertson et al. 2016) and observation of two professional AF match recordings identified 15 commonly performed player actions. Terminology and definitions of each player action are provided in Table 1. Notational software (Hudl Sportscode, v12.3.7) was used to manually label the player action of each performed acceleration and deceleration effort. A given effort could only be associated with one player action at a time. Intra-observer reliability was established by recoding the first match after a wash-out period of 5 weeks (Cohen's Kappa, K = 0.98). To assess inter-observer reliability, one match coded by the main match analyst was recoded by a second match analyst (K = 0.91).

Statistical analysis

Descriptive statistics, means and standard deviations, were calculated for each type of player action related to a high acceleration and/or deceleration effort.

Two linear mixed models (LMM) were conducted separately for high acceleration and deceleration efforts. The first LMM examined the commencing speed of players' high efforts across different player actions. The second LMM was conducted to examine how the commencing speed of players' high efforts affect the duration exceeding the high threshold of the effort. The LMMs included a random effect for player ID and game identity, which allows for different mean values for each player and game. The commencing speed of the first LMM model did not approximate a normal distribution and was log-transformed. Histograms and boxplots were created to show the distribution of the duration exceeding the high threshold and commencing speed of the high acceleration and deceleration efforts. All statistical analyses were performed using R (version 4.1.3) using the *lme4* package.

Results

Descriptive statistics of the high acceleration and deceleration efforts of each player action category can be found in Tables 2 and 3.

Duration exceeding the high threshold

Of all high efforts performed, 130 acceleration and 152 deceleration efforts exceeded the high threshold by a duration of 0.1 s, 58 acceleration and 156 deceleration by 0.2 s, 40 acceleration and 138 deceleration by 0.3 s, 10 acceleration and 98 deceleration by 0.4 s, 6 acceleration and 108 deceleration by 0.5 s, 8 acceleration and 86 deceleration by 0.6 s, 6 acceleration and 56 deceleration by 0.7 s, 10 acceleration and 46 deceleration by 0.8 s, 42 deceleration by 0.9 s, 6 deceleration by 1.0 s, 6 deceleration by 1.1 s, 6 deceleration by 1.2 s, and 4 deceleration efforts by 1.3 s. The duration exceeding the high threshold of high

Table 1. Definitions of commonly performed player actions in Australian football.

Player action	Definition
Body check	Deliberately obstructing a player by bumping into them.
COD <90	Change of direction less than 90 degrees from previous direction.
COD 90–180	Change of direction between 90–180 degrees from previous direction.
COD >180	Change of direction larger than 180 degrees from previous direction.
Dive	Jumping forwards (head and arms first) and falling on the ground to avoid something or to grab the ball.
Fall	Inadvertently lose balance and coming to rest on the ground.
Handball	Disposing of the football by hitting it with the clenched fist of one hand, whilst holding it with the other.
Jump	Push off the ground and into the air using legs.
Kick	Disposing of the football with any part of the leg below the knee.
Linear movement	Moving in a straight line.
Rotation	Rotation on the spot (in an area of 1 m^2)
Standing still	Remain in place.
Stop	Cease of motion.
Tackle	Taking hold of an opposition player.
TackleD	Taken hold off by an opposition player.

Table 2. Descriptive statistics (mean ± SD) of high *acceleration* efforts (≥3 $m \cdot s^{-2}$) categorised per player action.

Player action	Count (# pppg*)	Total duration (s)	Duration exceeding high threshold (s)	Average acceleration ($m \cdot s^{-2}$)	Max acceleration ($m \cdot s^{-2}$)	Commencing speed ($m \cdot s^{-1}$)	Average speed ($m \cdot s^{-1}$)	Max speed ($m \cdot s^{-1}$)
Linear movement	5 ± 2	1.63 ± 0.72	0.28 ± 0.21	1.94 ± 0.37	3.33 ± 0.27	3.57 ± 0.83	5.63 ± 0.90	5.63 ± 1.14
Jump	1 ± 1	1.51 ± 1.01	0.13 ± 0.06	2.04 ± 0.38	3.19 ± 0.17	3.12 ± 0.70	4.36 ± 1.27	5.16 ± 1.74
Kick	1 ± 1	1.20 ± 0.71	0.25 ± 0.06	2.18 ± 0.51	3.62 ± 0.45	3.36 ± 0.76	4.36 ± 1.14	4.99 ± 1.33
COD <90	1 ± 1	1.95 ± 0.89	0.16 ± 0.14	1.81 ± 0.39	3.25 ± 0.19	3.74 ± 0.86	5.24 ± 0.88	6.05 ± 1.09

*pppg is per player per game; max is maximum.

Table 3. Descriptive statistics (mean ± SD) of high *deceleration* efforts (≤3 m · s^{-2}) categorised per player action.

Player action	Count (# pppg*)	Total duration (s)	Duration exceeding high threshold (s)	Average acceleration (m · s^{-2})	Max acceleration (m · s^{-2})	Commencing speed (m · s^{-1})	Average speed (m · s^{-1})	Max speed (m · s^{-1})
Stop	10 ± 3	1.65 ± 2.14	0.42 ± 0.27	−1.96 ± 0.52	−3.61 ± 0.47	2.97 ± 1.01	1.6 ± 0.68	2.97 ± 1.01
TackleD	6 ± 2	1.91 ± 0.9	0.45 ± 0.29	−1.84 ± 0.50	−3.85 ± 0.74	2.99 ± 0.95	1.39 ± 0.59	3.02 ± 0.97
Tackle	4 ± 2	2.02 ± 0.85	0.47 ± 0.29	−2.02 ± 0.68	−3.91 ± 0.56	3.45 ± 0.98	1.64 ± 0.79	3.45 ± 0.98
Jump	4 ± 1	1.78 ± 1.17	0.38 ± 0.26	−1.92 ± 0.57	−3.65 ± 0.55	3.41 ± 1.09	2.01 ± 0.85	3.42 ± 1.08
COD <90	3 ± 1	1.47 ± 0.91	0.33 ± 0.19	−1.92 ± 0.48	−3.59 ± 0.60	2.95 ± 1.00	1.85 ± 0.71	2.96 ± 1.00
Body check	3 ± 1	1.72 ± 1.09	0.44 ± 0.29	−1.96 ± 0.59	−3.76 ± 0.63	3.33 ± 1.06	1.88 ± 0.75	3.35 ± 1.05
COD 90–180	2 ± 1	1.07 ± 0.48	0.38 ± 0.21	−2.28 ± 0.50	−3.63 ± 0.41	3.11 ± 0.93	2.14 ± 0.69	3.13 ± 0.90
COD >180	2 ± 1	1.27 ± 0.54	0.36 ± 0.22	−2.13 ± 0.60	−3.68 ± 0.62	3.46 ± 1.16	2.35 ± 0.84	3.46 ± 1.16
Fall	1 ± 1	1.35 ± 0.57	0.52 ± 0.28	−2.22 ± 0.51	−3.91 ± 0.55	3.27 ± 0.67	1.88 ± 0.71	3.27 ± 0.67
Dive	1 ± 1	1.81 ± 0.58	0.64 ± 0.29	−2.21 ± 0.40	−4.16 ± 0.73	3.34 ± 1.44	1.23 ± 0.44	3.34 ± 1.44
Handball	1 ± 0	0.90 ± 0.0	0.20 ± 0.00	−2.20 ± 0.20	−3.29 ± 0.31	3.58 ± 0.12	1.86 ± 0.32	2.58 ± 0.81
Kick	1 ± 0	2.32 ± 0.72	0.42 ± 0.32	−1.80 ± 0.50	−3.95 ± 0.97	3.47 ± 0.99	1.36 ± 0.43	3.47 ± 0.99

*pppg is per player per game; max is maximum.

acceleration and deceleration efforts for each type of player action are represented in Figure 1.

Commencing speed

The commencing speed of all high acceleration and deceleration efforts (all durations exceeding the high threshold combined) for each type of player action are represented in Figure 2. Of the high acceleration efforts, 41% were undertaken from a commencing speed between 3 and 4 m · s^{-1}, 31% from 4 to 5 m · s^{-1}, 17% from <3 m · s^{-1}, and 11% from ≥5 m · s^{-1}. For high deceleration efforts, 35% were undertaken from a commencing speed between 3 and 4 m · s^{-1}, 33% from 2 to 3 m · s^{-1}, 20% from 4 to 5 m · s^{-1}, 8% from ≥5 m · s^{-1}, and 5% from 1 to 2 m · s^{-1}. An overview of the different commencing speeds over durations exceeding the high threshold can be found in Figure 3.

No main effects were found in any of the acceleration efforts LMMs. In the first deceleration efforts LMM, there was a significant main effect for player action where the commencing speed of 'Jump' was significantly greater by 14% and 'Tackle' by 17% compared to the reference player action 'Stop' (Table 4). The second deceleration efforts LMM identified a significant main effect for commencing speed, indicating that for every unit increase in commencing speed (m · s^{-1}) the duration exceeding the deceleration threshold increased by 0.14 seconds (Table 5).

Discussion

The purpose of this study was to identify and profile the types of player actions of AF players executed during high acceleration and/or deceleration efforts and identify what duration exceeding the high threshold corresponds to these player actions. A secondary aim was to investigate the commencing speed profile of the acceleration and deceleration efforts. This study showed that 'linear movement' was the most frequently executed player action for high accelerations followed by 'jumps' (Figure 1(A)). The most frequently executed player actions for high deceleration were 'Stop' followed by 'TackleD' and 'Tackle' (Figure 1(B)). Differences were observed in commencing speed between player actions for the high deceleration efforts (Table 4). Furthermore, it was found that commencing speed influenced the duration exceeding the deceleration threshold (Table 5).

This study is the first to provide detailed information on the specific player actions executed during high acceleration and deceleration efforts. It has been observed that high

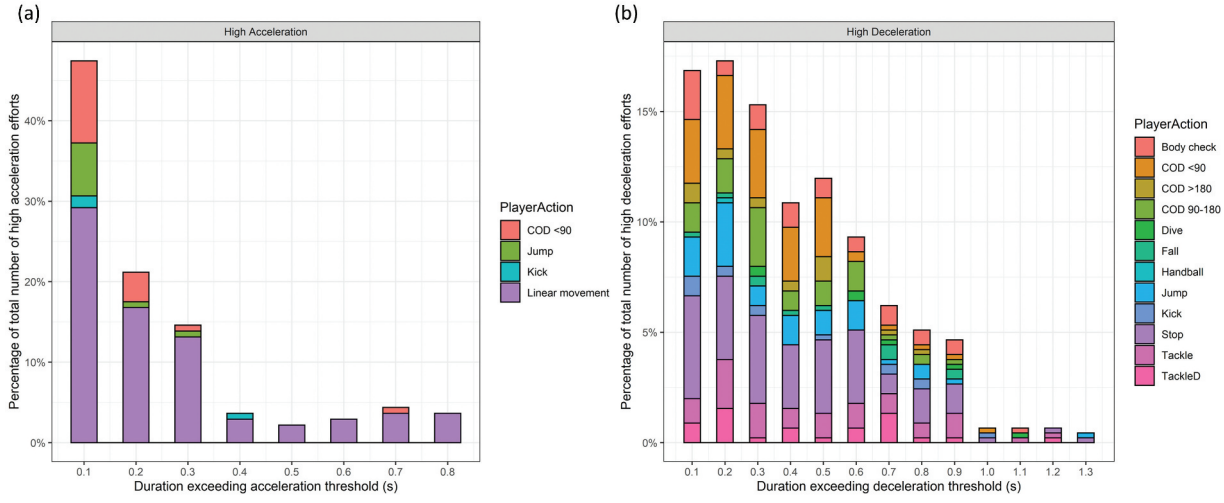

Figure 1. Histogram representing the duration exceeding the high threshold (±3 m · s^{-2}) of acceleration (A) and deceleration (B) efforts for each type of player action.

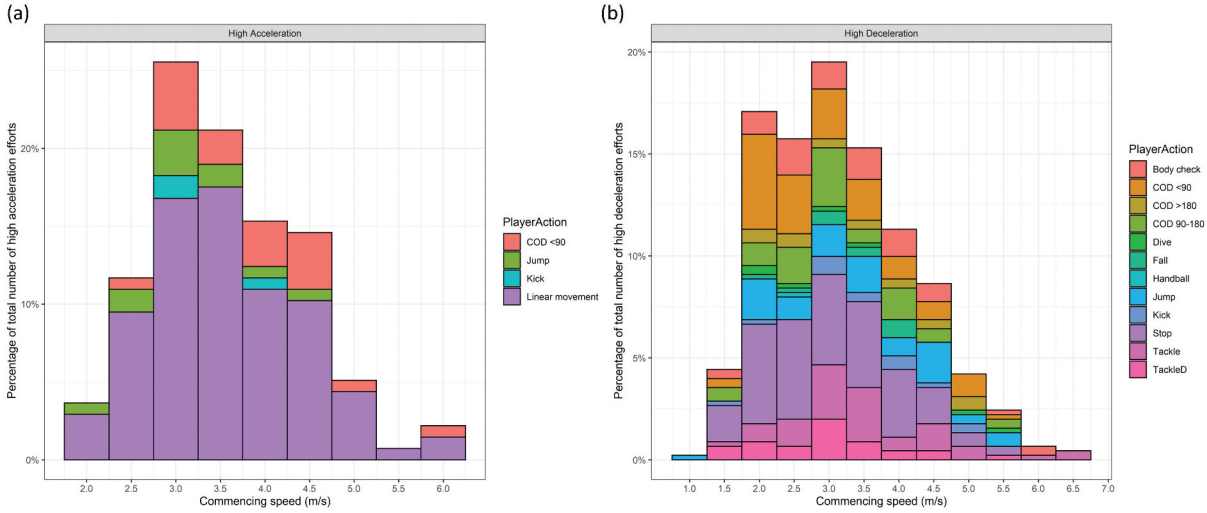

Figure 2. Histogram representing the commencing speed of high acceleration (a) and deceleration (b) efforts for each type of player action.

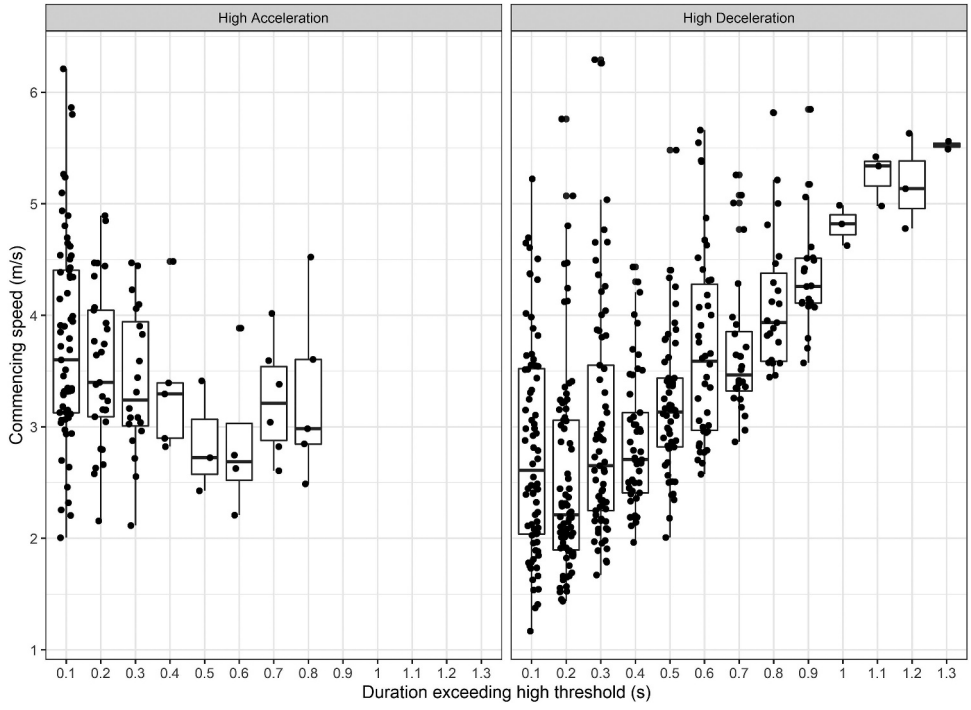

Figure 3. Box plots of the commencing speed of high acceleration (left panel) and high deceleration (right panel) efforts as a function of the duration exceeding the high threshold ($\pm 3\ m \cdot s^{-2}$).

decelerations occur more frequently than high accelerations, which is representative of those reported in the literature for AF (Gastin et al. 2019; Kelly et al. 2019). Presumably, this difference is due to competitive match play (Harper and Kiely 2018), where accelerations may occur more gradually, in a linear fashion, without reaching the set high acceleration threshold. This is supported by the finding that linear movements were the most frequently occurring player action for high accelerations in this study. High deceleration efforts, on the other hand, have a larger proportion of incidental efforts (Harper and Kiely 2018), which is also in line with findings in this study that a large proportion of deceleration efforts were related to incidental player actions, such as getting tackled, resulting in more frequently occurring efforts.

Different durations exceeding the high threshold were associated with different player actions, where the majority of high acceleration and deceleration efforts occurred with lower durations exceeding the high threshold (≤0.3 s) (Figure 1), which is in line with previous research (Varley et al. 2017). The efforts associated with lower durations exceeding the high threshold indicate that the player actions were the product of a spike in the acceleration, most likely related to short explosive actions. This can also be seen in the results where the short accelerating explosive action of

Table 4. Linear mixed model fixed and random effect parameters detailing the percentage mean difference from the intercept value of the speed ($m \cdot s^{-1}$) players typically commence their high effort from across different player actions. The fixed effect intercept term and random effects have been back transformed into their raw units from log units.

	Commencing speed ($m \cdot s^{-1}$)			
Fixed effects	Estimates	Lower 95% CI	Upper 95% CI	P
Intercept	3.03 $m \cdot s^{-1}$	2.83	3.22	<.001
Body check [a]	11.63%	−1.00	24.61	0.066
COD 90–180 [a]	4.08%	−6.76	16.18	0.506
COD <90 [a]	−1.98%	−10.42	8.33	0.699
COD >180 [a]	15.03%	−2.96	34.99	0.099
Dive [a]	8.33%	−14.79	39.10	0.511
Fall [a]	12.75%	−7.69	37.71	0.241
Handball [a]	−10.42%	−52.29	68.20	0.722
Jump [a]	13.88%	2.02	27.12	0.022
Kick [a]	17.35%	−1.00	40.49	0.073
Tackle [a]	17.35%	5.13	31.00	0.003
TackleD [a]	−1.00%	−13.06	13.88	0.934
Random effects				
Between Subject SD	1.04			
Between Game SD	1.01			
Residual SD	1.40			
ICC	0.02			

Note: [a] compared to player action 'Stop'.

Table 5. Linear mixed model fixed and random effect parameters detailing how the speed ($m \cdot s^{-1}$) players commence their high deceleration efforts at affect the duration(s) exceeding the high threshold ($\leq -3\ m \cdot s^{-1}$) of the effort.

Duration exceeding deceleration threshold (seconds)				
Fixed effects	Estimates	Lower 95% CI	Upper 95% CI	P
Intercept	0.41 s	0.38	0.43	<.001
Commencing speed	0.14 s	0.12	0.16	<.001
Random effects				
Between Subject SD	0.02			
Between Game SD	0.01			
Residual SD	0.25			
ICC	0.02			

jumping has only been identified with durations exceeding the high threshold by ≤0.3 s. Other player actions only appear until a certain duration exceeded the high threshold, e.g., decelerating change of directions above 180° mostly occurred with durations exceeding the high threshold by 0.1–0.5 s and rarely appeared above 0.5 s. Choosing an appropriate minimum duration above the threshold for detecting high acceleration and deceleration efforts is an important consideration for researchers and practitioners, as it will determine what player actions comprise the detected efforts. For example, if a practitioner is mainly interested in detecting high acceleration efforts of a linear nature (linear movement), a minimum duration above the threshold (dwell time) of ≥0.5 s (duration exceeding the high threshold by 0.5 s, see Figure 1) would best resemble the linear movement player action and exclude movements they are not interested in.

It was found that commencing speed influenced the duration exceeding the deceleration threshold, where a higher commencing speed is related to a larger duration exceeding the deceleration threshold (Table 5). This indicates that when a player is running at a higher speed and needs to slow down, they will require a high magnitude of deceleration and will therefore spend a greater amount of time above the deceleration threshold. This difference was not found within the acceleration efforts, which may indicate that the speed needed to cross the high acceleration threshold of $3\ m \cdot s^{-1}$ are accumulated over a variety of commencing speeds (Morin et al. 2021). These results indicate that the ability to accelerate and decelerate is highly dependent on the commencing speed of the player, particularly when decelerating.

With high acceleration and deceleration efforts being linked to elevated muscle damage after AF competition (Gastin et al. 2019), it is crucial that acceleration and deceleration movements are appropriately trained. Our results suggest that AF players mostly accumulate high acceleration efforts by linear movements, and high deceleration efforts by stop of motion, physical contact actions, such as tackling, and changing of direction. It has been shown that the player actions corresponding to the high acceleration and deceleration efforts occur from commencing speeds ranging from 1 to $6\ m \cdot s^{-1}$, where the commencing speed of the player actions jump and tackling of the high deceleration effort were significantly higher from the player action stop. To properly prepare AF players for competition, the enhancement of accelerating linear movements, decelerating physical contact and stop actions, over a variety of commencing speeds may be beneficial during AF games.

Instead of simply reporting the frequency of the high acceleration and deceleration, this study presents specific player actions and their speed profile as a characteristic signature that differentiates high acceleration and deceleration efforts by player action. Potential applications may include a more thorough representation of the acceleration and deceleration effort variable than can be learned by using the frequency alone. It is recommended that training practices replicate the player actions over various commencing speeds evident in this research, for high acceleration and deceleration efforts.

Using an absolute threshold ($\pm 3\ m \cdot s^{-2}$) for high efforts allowed comparison of the results across different studies. However, it did not account for variation in individual player capacities and may have provided an inaccurate representation of each player's true intensity of acceleration (Abbott et al. 2018). Players with different physical capacities may execute different player actions to achieve the absolute threshold, the actions might change if a relative threshold was to be used.

A noteworthy strength of this investigation was the use of elite AF team data, where players are exposed to a homogeneous training environment, which may result in more consistent player actions. As with all match analysis research, the findings of this research are representative of AF at the time of publication, as the dynamic nature of team sport means that the player movement will change over time. Furthermore, the data used were of AF players only and may not be representative of other team sports. Future research should investigate the player actions and their frequency that correspond to high acceleration and deceleration efforts across a range of different team sports, this to provide a thorough evaluation of the acceleration and deceleration effort variable across team sports.

Practical application

- Practitioners and researchers should be aware that selecting a minimum-duration above the threshold for detecting high acceleration and deceleration efforts will influence the corresponding type of player actions detected during these efforts.
- Practitioners and researchers can use the findings of this research to help with selecting minimum-duration criteria that best resemble the player's actions of interest and exclude movements they are not interested in.
- It is recommended that training practices for high acceleration and deceleration efforts replicate the player actions associated with these efforts evident in this research, over various commencing speeds.

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